

Kirk Tolleshaug

Senior Backend Software Engineer | C#.NET | APIs, Distributed Systems & Architecture

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Summary

I'm a senior software engineer focused on backend systems, APIs, architecture, and performance.

My technical career started in high school with hands-on IT support and deployment work and grew into 12+ years of professional software development across C#.NET, REST APIs, distributed and event-driven systems, relational databases, cloud-hosted applications, and legacy modernization.

Recent work has focused heavily on .NET and SQL Server backend services supporting web, mobile, external integrations, and high-volume event processing across systems serving tens of thousands of users and millions of devices.

I've improved API response times from several seconds to sub-second performance, reduced long-running processing from hours to minutes and minutes to seconds, modernized legacy .NET services, and built event-driven and parallel processing workflows.

I've also contributed through architecture ownership, technical design, project decomposition, mentoring, and end-to-end delivery.

I'm primarily interested in senior backend roles with opportunities to stay hands-on while contributing to architecture, technical direction, and modernization.

Technical Skills

Languages & Frameworks: C#, .NET, ASP.NET, Go, Java, Angular, React

Backend & Architecture: REST APIs, Distributed Systems, Event-Driven Architecture, gRPC, Background Services, Multithreading, Parallel Processing, Technical Leadership

Data & Messaging: Microsoft SQL Server, PostgreSQL, Entity Framework, RabbitMQ, Redis, Distributed Caching

Cloud & Delivery: Azure, AWS, Infrastructure as Code, CI/CD, Azure App Services, Azure SQL, Key Vault, Azure Functions, IIS

Professional Experience

Digital Monitoring Products

Software Engineer · December 2022 - Present · Springfield, MO

Work across backend services, APIs, distributed processing, device integrations, delivery automation, and web applications.

- Design and optimize C#.NET backend services and REST APIs supporting internal web and mobile applications, external integrations, tens of thousands of users, and millions of security devices.
- Diagnose performance bottlenecks involving memory, request processing, concurrency, and workload distribution, reducing some operations from hours to minutes, others from minutes to seconds, and select API paths from several seconds to sub-second response times.
- Build event driven and distributed workflows using RabbitMQ, background services, Redis, distributed caching, and parallel processing for device events and job coordination.
- Modernize legacy services from .NET Framework 4.x and .NET Core 3 to modern .NET, while improving CI/CD automation and deployment reliability.
- Design SQL queries, stored procedures, and indexes in both MS SQL Server and PostgreSQL relational and time-series databases.
- Developed React based user interfaces and application workflows.
- Designed a Go based integration running directly on security cameras, using REST APIs and bidirectional gRPC for configuration, management, and delivery of events triggered by on camera AI detection.
- Prototyped mesh communication allowing cameras to relay messages when direct connectivity was unavailable.
- Provide technical leadership through architecture, design discussions, project decomposition, mentoring, and coordination across engineering and QA.

Vizient, Inc

Senior Software Engineer · June 2014 - December 2022 · Cape Girardeau, MO

Progressed from software engineering intern to senior software engineer over eight years, taking on increasing responsibility for architecture, technical ownership, team leadership, and end to end delivery.

Worked across identity/SSO, reporting, legacy modernization, and Azure hosted applications.

- Led technical design and delivery across multiple initiatives, decomposing complex work, mentoring engineers, coordinating with QA, and helping set implementation direction for the team.
- Architected and owned specialized projects end to end, serving as the primary technical owner for requirements, system design, implementation, deployment, and ongoing support.
- Built cloud spend and usage reporting that identified unnecessary services and optimization opportunities, contributing to approximately \$30M in realized annual infrastructure savings.
- Designed and developed C#.NET applications and backend services across multiple generations of .NET, with Angular front ends and SQL Server-backed data stores.

- Improved backend and API performance, reducing select response times from several seconds to sub-second performance.
- Built Azure hosted applications using App Services, Azure SQL, Key Vault, Functions, and infrastructure as code, including Pulumi and early Bicep adoption.
- Developed reusable infrastructure components and CI/CD automation to standardize application provisioning and deployment.
- Developed and supported OAuth/SSO integrations and user management capabilities across multiple applications and integration scenarios.
- Designed and delivered a complex reporting system that combined data across multiple domains and schemas, performed extensive transformation and customer specific formatting, and processed datasets containing millions of records.

Earlier Experience

Southeast Missouri State University

Teaching Assistant · January 2015 - December 2015 · Cape Girardeau, MO

- Assisted faculty with Computer Science I, II, III, C and Unix, and Logic Fundamentals courses.
- Helped students with programming concepts, debugging, coursework, and technical problem solving.
- Graded assignments and coursework, provided feedback to students, and assisted with exam proctoring.

Southeast Missouri State University

WebParaprofessional · January 2014 - May 2015 · Cape Girardeau, MO

- Designed and implemented ASP.NET/C# web applications for university departments using SQL Server, Entity Framework, and IIS.
- Designed and built an end to end sign-in and usage-tracking application for the university writing lab.
- Supported Moodle and related instructional systems for faculty and staff.
- Produced instructional video and voice content supporting university technology and course delivery.

Education & Certification

Southeast Missouri State University · Bachelor of Science - BS, General Studies · 2011 - 2015

Azure Fundamentals